

OmniCentricBot Knowledge Brief

Version: 2.4.1

Published: 12 March 2025

Author: Dr. Lena Ortiz

Organization: Northwind Applied AI Lab

1. Mission

OmniCentricBot is a context-grounded multimodal RAG assistant. It answers questions only from uploaded documents and refuses when evidence is missing.

2. Core Numbers

- Embedding model dimension: 384
- Default relevance threshold: 0.35
- Maximum upload size: 50 MB
- Whisper free-tier media limit: 25 MB
- Supported image formats: PNG, JPG, WEBP, GIF, BMP

3. Architecture Overview

Retrieval uses hybrid search: dense vectors in Qdrant plus BM25 keyword matching, fused with Reciprocal Rank Fusion (RRF), then a cross-encoder reranker (Xenova/ms-marco-MiniLM-L-6-v2). Generation defaults to Groq model openai/gpt-oss-20b with Gemini as fallback.

4. Safety Policy

The assistant must not invent phone numbers, home addresses, or medical dosages that are absent from the source documents. If asked for the capital of France and that fact is not in the documents, it must refuse.

5. Project Codename

Internal codename: Project Harborlight

Primary contact email: harborlight@northwind-lab.example

Launch city: Portland